Assignment-3

1. Create a Flask application with an /api route. When this route is accessed, it should return a JSON list. The data should be stored in a backend file, read from it, and sent as a response.

app_new.py > ...

```python
from flask import Flask, jsonify
import json
import os

app = Flask(__name__)

# Path to the data file
DATA_FILE = os.path.join(os.path.dirname(__file__), 'data.json')

@app.route('/')
def home():
    return jsonify({'message': 'Welcome to Flask API'})

@app.route('/api')
def api():
    """Return JSON data from data.json file"""
    try:
        with open(DATA_FILE, 'r') as file:
            data = json.load(file)
        return jsonify(data)
    except FileNotFoundError:
        return jsonify({'error': 'Data file not found'}), 404
    except json.JSONDecodeError:
        return jsonify({'error': 'Invalid JSON in data file'}), 500

if __name__ == '__main__':
    app.run(debug=True)

```

Objective

This program creates a simple REST API using the Flask framework. It provides two endpoints:

1. **Home Route (/)** – Displays a welcome message.
2. **API Route (**/api**)** – Reads data from a JSON file and returns it as a JSON response.

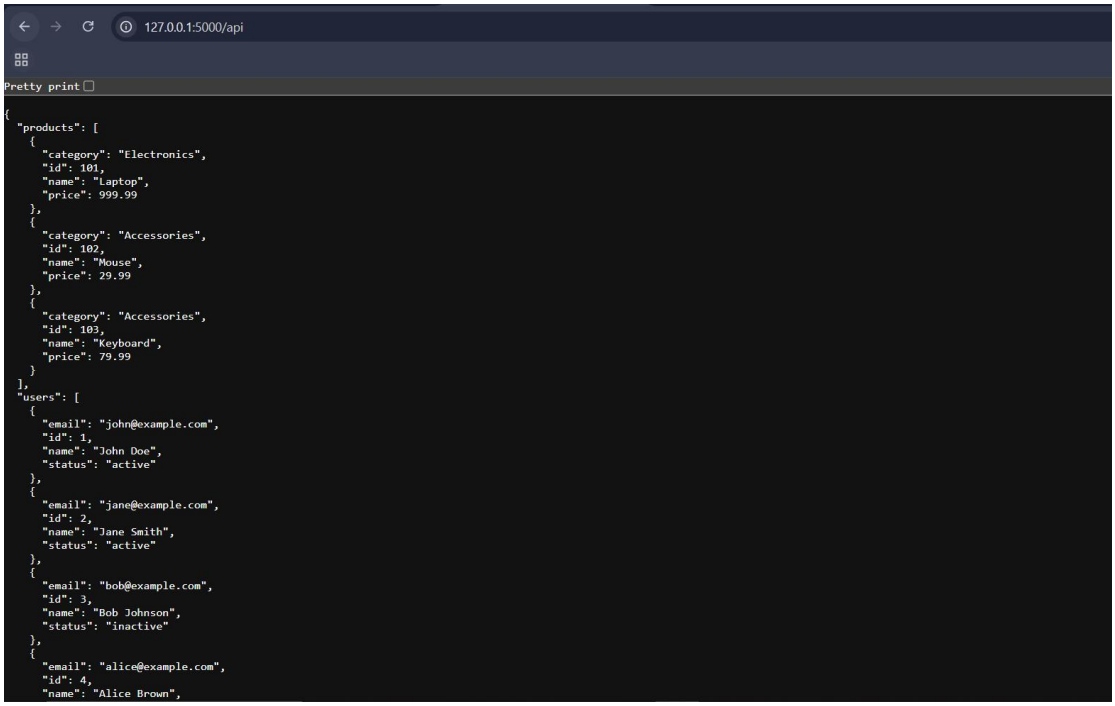
Data.json


{} data.json > [ ] products > { } 2
{
"users": [
{
"id": 1,
"name": "John Doe",
"email": "john@example.com",
"status": "active"
},
{
"id": 2,
"name": "Jane Smith",
"email": "jane@example.com",
"status": "active"
},
{
"id": 3,
"name": "Bob Johnson",
"email": "bob@example.com",
"status": "inactive"
},
{
"id": 4,
"name": "Alice Brown",
"email": "alice@example.com",
"status": "active"
}
],
"products": [
{
"id": 101,
"name": "Laptop",
"price": 999.99,
"category": "Electronics"


],
"products": [
{
"id": 101,
"name": "Laptop",
"price": 999.99,
"category": "Electronics"
},
{
"id": 102,
"name": "Mouse",
"price": 29.99,
"category": "Accessories"
},
{
"id": 103,
"name": "Keyboard",
"price": 79.99,
"category": "Accessories"
}
]
}

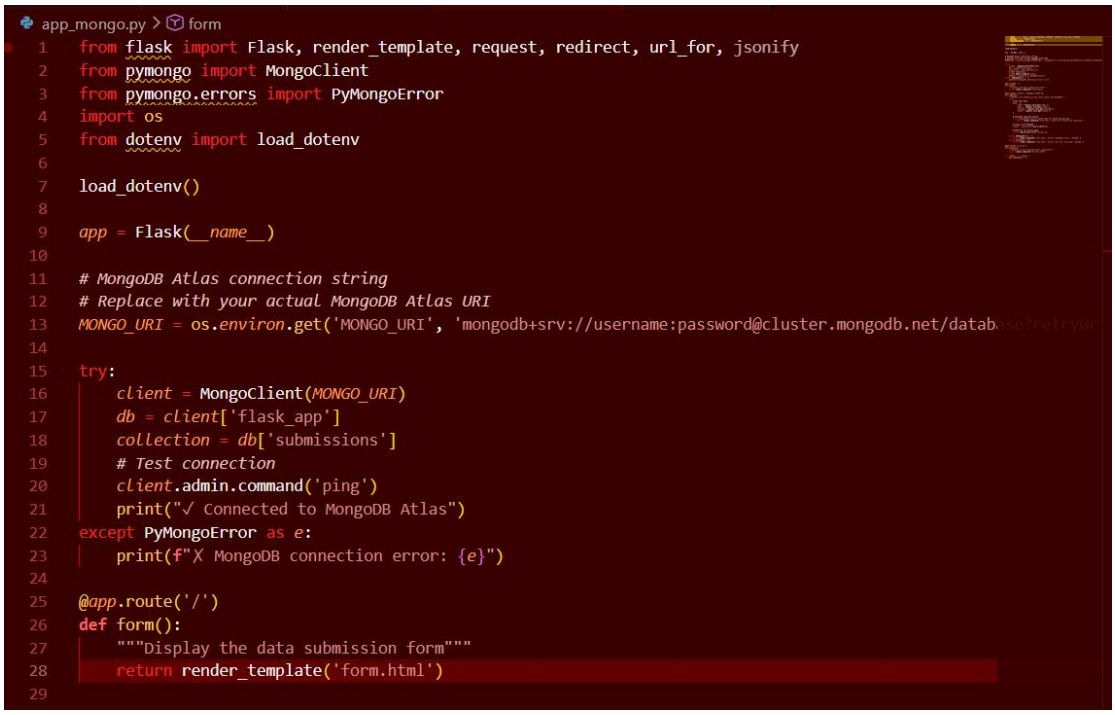
Home Route (http://localhost:5000/)

API Route (http://localhost:5000/api)



2.Create a form on the frontend that, when submitted, inserts data into MongoDB Atlas. Upon successful submission, the user should be redirected to another page displaying the message **"Data submitted successfully"**. If there's an error during submission, display the error on the same page without redirection.

app_mongo.py



```python
app_mongo.py > form
from flask import Flask, render_template, request, redirect, url_for, jsonify
from pymongo import MongoClient
from pymongo.errors import PyMongoError
import os
from dotenv import load_dotenv

load_dotenv()

app = Flask(__name__)

# MongoDB Atlas connection string
# Replace with your actual MongoDB Atlas URI
MONGO_URI = os.environ.get('MONGO_URI', 'mongodb+srv://username:password@cluster.mongodb.net/datab
try:
    client = MongoClient(MONGO_URI)
    db = client['flask_app']
    collection = db['submissions']
    # Test connection
    client.admin.command('ping')
    print("✓ Connected to MongoDB Atlas")
except PyMongoError as e:
    print(f"X MongoDB connection error: {e}")

@app.route('/')
def form():
    """Display the data submission form"""
    return render_template('form.html')
```

```python
@app.route('/submit', methods=['POST'])
def submit():
    """Handle form submission and insert data into MongoDB"""
    try:
        # Get form data
        data = {
            'name': request.form.get('name'),
            'email': request.form.get('email'),
            'message': request.form.get('message'),
            'phone': request.form.get('phone')
        }

        # Validate required fields
        if not all([data['name'], data['email'], data['message']]):
            return render_template('form.html', error='All fields are required!')

        # Insert into MongoDB
        result = collection.insert_one(data)

        # Redirect to success page
        return redirect(url_for('success'))

    except PyMongoError as e:
        return render_template('form.html', error=f'Database error: {str(e)}')
    except Exception as e:
        return render_template('form.html', error=f'An error occurred: {str(e)}')
```

```python

@app.route('/success')
def success():
    """Display success message after submission"""
    return render_template('success.html')

if __name__ == '__main__':
    app.run(debug=True)

```

Objective

This program is a Flask web application that allows users to submit data through an HTML form and stores the submitted information in a MongoDB Atlas database. It also validates user input and displays a success page after successful submission.

Success.html

templates > success.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Submission Successful</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            min-height: 100vh;
            display: flex;
            justify-content: center;
            align-items: center;
            padding: 20px;
        }

        .container {
            background: white;
            border-radius: 10px;
            box-shadow: 0 10px 40px rgba(0, 0, 0, 0.2);
            padding: 60px 40px;
            max-width: 500px;
            width: 100%;
            text-align: center;
        }
```

templates > success.html > ...

```
<html lang="en">
<head>
    <style>

        .success-icon {
            width: 80px;
            height: 80px;
            background: #4caf50;
            border-radius: 50%;
            display: flex;
            justify-content: center;
            align-items: center;
            margin: 0 auto 30px;
            animation: scaleIn 0.5s ease-out;
        }

        .success-icon::after {
            content: '✓';
            color: white;
            font-size: 48px;
            font-weight: bold;
        }

        @keyframes scaleIn {
            from {
                transform: scale(0);
            }
            to {
                transform: scale(1);
            }
        }

```

templates > success.html > ...

```
<html lang="en">
<head>
    <style>
        @keyframes scaleIn {
        }

        h1 {
            color: #333;
            font-size: 32px;
            margin-bottom: 15px;
        }

        .message {
            color: #666;
            font-size: 16px;
            margin-bottom: 30px;
            line-height: 1.6;
        }

        .details {
            background: #f5f5f5;
            padding: 20px;
            border-radius: 5px;
            margin-bottom: 30px;
            text-align: left;
        }

        .details p {
            color: #555;
            font-size: 14px;
            margin-bottom: 8px;
        }
```

templates > success.html > ...

```
 2  <html lang="en">
 3  <head>
 7      <style>
83          .details p {
87          }
88
89          .details strong {
90              color: #333;
91          }
92
93          .button-group {
94              display: flex;
95              gap: 10px;
96          }
97
98          a,
99          button {
100             flex: 1;
101             padding: 12px;
102             border: none;
103             border-radius: 5px;
104             font-size: 16px;
105             font-weight: 600;
106             cursor: pointer;
107             text-decoration: none;
108             transition: transform 0.2s, box-shadow 0.2s;
109             display: inline-block;
110         }
111
```

templates > success.html > ...

```
 2  <html lang="en">
 3  <head>
 7      <style>
98          a,
110         }
111
112         .btn-primary {
113             background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
114             color: white;
115         }
116
117         .btn-primary:hover {
118             transform: translateY(-2px);
119             box-shadow: 0 10px 20px rgba(102, 126, 234, 0.3);
120         }
121
122         .btn-secondary {
123             background: #f0f0f0;
124             color: #333;
125             border: 2px solid #e0e0e0;
126         }
127
128         .btn-secondary:hover {
129             background: #e8e8e8;
130             border-color: #d0d0d0;
131         }
```

templates > success.html > ...

```
<html lang="en">
<head>
    <style>
        .btn-secondary:nover {
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="success-icon"></div>

        <h1>Data Submitted Successfully!</h1>

        <p class="message">
            Your information has been received and stored in our database.
            We'll review your submission and get back to you shortly.
        </p>

        <div class="details">
            <p><strong>Status:</strong> ✓ Submission Completed</p>
            <p><strong>Date & Time:</strong> <span id="timestamp"></span></p>
            <p><strong>Message:</strong> Thank you for your submission</p>
        </div>

        <div class="button-group">
            <a href="{{ url_for('form') }}" class="btn-primary">Submit Another</a>
            <a href="/" class="btn-secondary">Go Home</a>
        </div>
    </div>

```

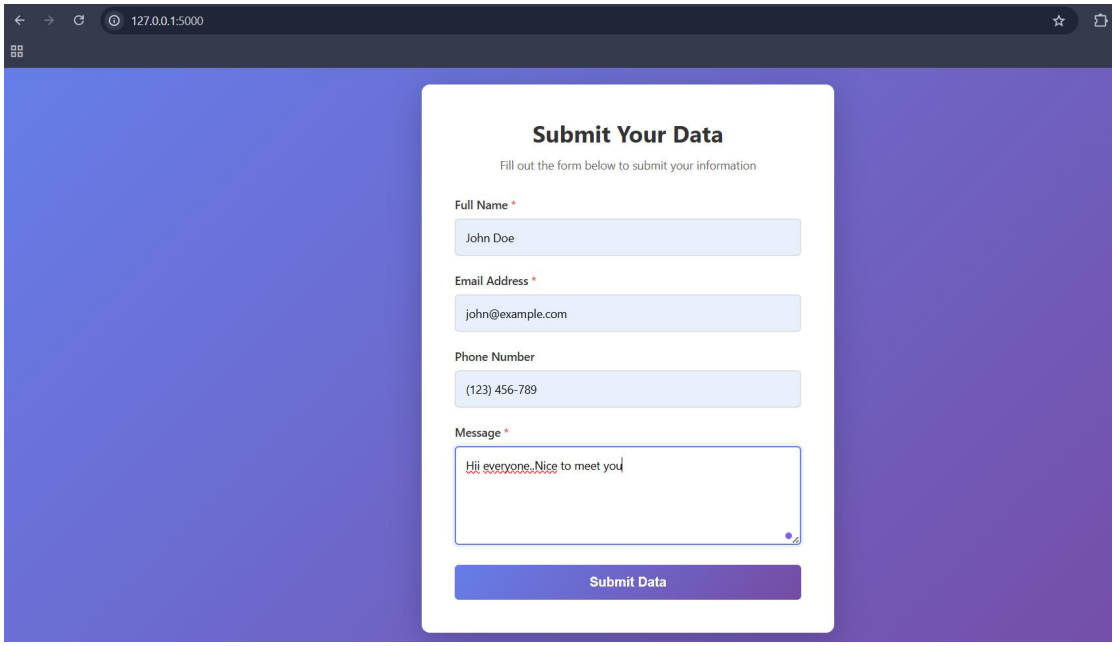
templates > success.html > ...

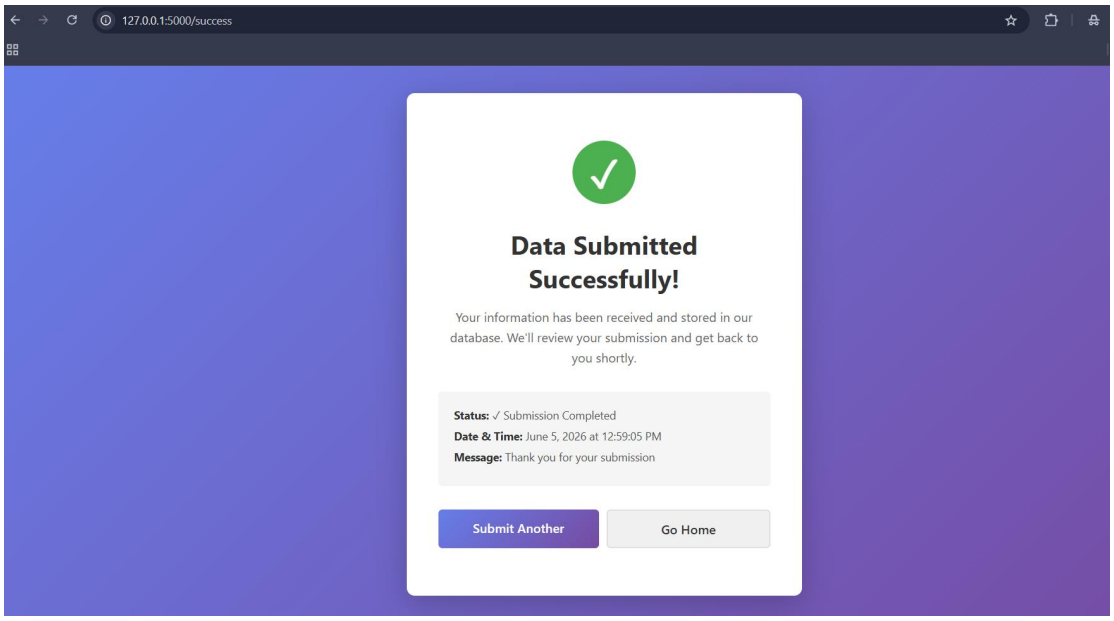
```
<html lang="en">
<body>
    <div class="container">
        </div>

        <div class="button-group">
            <a href="{{ url_for('form') }}" class="btn-primary">Submit Another</a>
            <a href="/" class="btn-secondary">Go Home</a>
        </div>
    </div>

    <script>
        // Display current date and time
        const now = new Date();
        const options = {
            year: 'numeric',
            month: 'long',
            day: 'numeric',
            hour: '2-digit',
            minute: '2-digit',
            second: '2-digit'
        };
        document.getElementById('timestamp').textContent = now.toLocaleDateString('en-US', options);
    </script>
</body>
</html>

```


127.0.0.1:5000
Submit Your Data
Fill out the form below to submit your information
Full Name *
John Doe
Email Address *
john@example.com
Phone Number
(123) 456-789
Message *
Hii everyone..Nice to meet you
Submit Data


127.0.0.1:5000/success
Data Submitted Successfully!
Your information has been received and stored in our database. We'll review your submission and get back to you shortly.
Status: ✓ Submission Completed
Date & Time: June 5, 2026 at 12:59:05 PM
Message: Thank you for your submission
Submit Another
Go Home